

Continuous functions – 4

Uniform continuity and the modulus of continuity

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Theoretical minimum.

Def. f is *uniformly continuous* on A if for every $\varepsilon > 0$ there is $\delta > 0$ with $|f(x) - f(y)| < \varepsilon$ for all $x, y \in A$ with $|x - y| < \delta$. The *modulus of continuity* is $\omega_f(\delta) = \sup\{|f(x) - f(y)| : |x - y| \leq \delta\}$; uniform continuity means exactly $\omega_f(\delta) \rightarrow 0$ as $\delta \rightarrow 0$.

Theorem 1 (Cantor). A continuous function on a compact set is uniformly continuous. Consequently a continuous function on a bounded set is uniformly continuous if and only if it extends continuously to the closure.

Theorem 2 (linear growth). A uniformly continuous $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies $|f(x)| \leq a|x| + b$ for some constants $a, b > 0$; hence x^2 , e^x and $\sin(x^2)$ are not uniformly continuous on the line, while $\sqrt{x}$ is uniformly continuous on $[0, \infty)$ despite its unbounded derivative.

Fact 1. f is uniformly continuous on A if and only if $f(x_n) - f(y_n) \rightarrow 0$ for every pair of sequences with $x_n - y_n \rightarrow 0$; this sequential form is what one actually uses to disprove uniform continuity.

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1. Prove Cantor's theorem, and prove that a uniformly continuous function on a bounded set extends continuously to its closure. Deduce that $\sin \frac{1}{x}$ is not uniformly continuous on $(0, 1)$.
 2. (Iranian competition) Prove that a uniformly continuous $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies $|f(x)| \leq a|x| + b$ for suitable positive a and b ; deduce that no uniformly continuous function on $\mathbb{R}$ grows faster than linearly.
 3. Decide which of the following are uniformly continuous on the indicated sets and justify each answer through the modulus of continuity: $\sqrt{x}$ on $[0, \infty)$; x^2 on $\mathbb{R}$ and on $[0, 1]$; $\sin(x^2)$ on $\mathbb{R}$; $x \sin x$ on $\mathbb{R}$; $\frac{\sin x}{x}$ on $(0, \infty)$.

4. (S.T. Yau CSMC 2018) Let $A \subseteq \mathbb{R}^m$ and let $f: A \rightarrow \mathbb{R}$ be L -Lipschitz. Prove that f extends to an L -Lipschitz function on all of $\mathbb{R}^m$ (McShane's extension).
5. (Iranian competition) Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be continuous and such that $\lim_{x \rightarrow +\infty} (g(x+t) - g(x)) = 0$ for every $t \in \mathbb{R}$. Prove that for every compact $K \subset \mathbb{R}$ the convergence is uniform in $t \in K$.